Coventry University

School of Engineering

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7011CEM: Individual Project

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Project Title: Analyzing of Motor Characteristics for Shell ECO Marathon Project

Course: MSc Electrical and Electronics Engineering

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Declaration

	I submitted my ethics application, and my application has been approved. I include my ethics certificate in the appendix as evidence
	I submitted my ethics application, and my application is currently under review
☑	**I have not submitted my ethics application**

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Research Question, Problem Statement or Topic for Investigation

1.1 Project topic:

Analyzing of Motor Characteristics for Shell ECO Marathon Project.

1.2 Research Aim:

The aim of this project is to build an experimental setup and apply the up-to-date experimental tests on the project PMSM to identify its parameters.

1.3 Research Question:

How can accurate PMSM parameter identification contribute to improving the design, tuning, and performance of motor control algorithms in electric vehicle applications?

1.4 problem statement:

Precise control of electric machine drives requires some complicated control algorithms such as the Direct torque control and Field oriented control which provide precise speed and torque control. These types of control are based on the machine modeling and thus they are relying on the identification of the machine parameters. Applying this control strategy for the motor that will be used in the development of the electric power train for the Eco Shell Marathon proto vehicle within the school of engineering is implying the need for precise identification of the PMSM that will be used. The aim of this project is to build an experimental setup and apply the up-to-date experimental tests on the project PMSM to identify its parameters. Thus, the purpose of this study is to examine how precise PMSM parameter identification can improve the performance and design of FOC-based drive systems, especially for electric vehicle applications.

2. Intended User or Group of Users and Their Requirements

The intended audience of the proposed project consists mainly of the engineering professionals, researchers, and students majoring in electrical drives, power electronics, and control systems. It is also of great importance to the industrial sectors, including the electric vehicle propulsion, robotics, and automated manufacturing, where the optimization of performance and reliability of a system requires a high efficiency and accuracy of motor speed control.

3. Systems Requirements, Project Deliverables, and Final Project Outcome

The project involves the design and development of an **experimental setup** for the characterization and control of a **Permanent Magnet Synchronous Motor (PMSM)**. The system will be configured to perform detailed parameter identification tests on a **350 W PMSM**, which will later support the implementation of a **Field-Oriented Control (FOC)** strategy for high-performance speed regulation.

Stages of the Project and Expected Deliverables at Each Stage:
The key requirements of the system are:

- A 350 W Permanent magnet Synchronous Motor (PMSM).
- Hardware Experimental setup including the fixation of the motor, braking system to control the motor loading, for load testing, multi-meters with suitable voltage and current measurement range, oscilloscopes, etc.
- Measurement devices also include current and voltage sensors, torque, and speed sensors, and data acquisition devices.
- Microcontroller OR DSP platform for implementing the FOC algorithm.
- Safety and protection for stable and reliable operations.

The expected project deliverables are as follows:

- Entire experimental that can be used to identify the parameters of PMSM.
- Effective characterization of the PMSM by experimental means, which produces correct establishment of:
    - Direct- and quadrature-axis inductances (Ld and Lq)
    - Stator winding resistance (Rs)
    - Permanent-magnet flux linkage
    - Rotor inertia constant (J)
- A verified dataset containing the measured electrical and mechanical parameters of the PMSM.
- Implementation of these parameters in a **Field-Oriented Control (FOC)** algorithm using a microcontroller and data acquisition system.

Ultimate Goal of the Project:

Applying these parameters to a FOC of the motor using microcontroller and data acquisition system for precise control of the motor speed and torque for the application in the electric power train of the Shell Eco Marathon Proto vehicle.

4. Primary Research Plan

Task / Activity	Description	Duration (Weeks)
Literature review	Research on the working principle of PMSM and how it is used in electric vehicles. Academic papers on parameter identification and Field-Oriented Control (FOC).	Weeks: 1-2
Preparation of Bill of Materials (BoM)	Prepare a list of all the necessary equipment, components, devices that needed to be tested in the experiment (inverter, sensors, DAQ, motor, microcontroller).	Week: 2-3
Construction of Experimental Setup	Connect the sensors, inverter, and data acquisition system to the PMSM test bench. Verify safety and calibration compliance.	Week: 3-5
Experimental Testing	Measure the PMSM's electrical and mechanical parameters using locked-rotor and no-load tests.	Week: 5-7
Data Analysis and Parameter Calculation	Calculate Rs, Ld, Lq, Psf, and J through evaluating the data that was gathered. Use theoretical models to validate the results.	Week: 7-9
Documentation and report presentation	Compile the results, statistics, and analysis into the presentation slides and project dissertation.	Week: 10-11
Collaboration with SEM Team	Contribute identified parameters and control results to the Shell Eco Marathon prototype vehicle team for real-world application.	Week: 12

5. Initial Literature Review

The interaction between the rotating magnetic field of the stator and the steady magnetic field produced by permanent magnets in the rotor is the basis for the operation of permanent magnet synchronous motors, or PMSMs. PMSMs are more reliable, more efficient, and have a faster dynamic response than traditional induction machines, which makes them ideal for sophisticated drive systems (Krishnan, 2010).

Since it can decouple torque and flux components through coordinate transformation, which is determined by the rotor flux angle, Field-Oriented Control (FOC) is the most widely used control strategy (Zhu and Howe, 2007).

This method preserves high precision and smooth torque output while reducing the controller's computational load. FOC is perfect for industrial motor control applications and electric vehicle applications because it also allows for higher Pulse Width Modulation (PWM) frequencies and enhanced system responsiveness (Lee et al., 2020).

Recent research thus emphasizes the continuous requirement for hardware validation of PMSM models and experimental parameter identification in order to support dependable control performance and fault diagnosis in actual drive systems (Combes et al., 2017; Bianchi et al., 2017).

A number of experimental methods for identifying PMSM parameters have been proposed in recent studies to increase the precision of dynamic modelling and Field-Oriented Control (FOC). Key parameters, including stator resistance, are frequently determined using conventional techniques like the DC resistance test, locked-rotor impedance measurement, and no-load back-EMF testing (R_s), inductances (L_d, L_q), and permanent magnet flux linkage (ψ_f) (Lee *et al.*, 2020).Flux linkage and incremental inductances under saturation and cross-coupling conditions have been demonstrated to be more accurately estimated using advanced approaches such as high-frequency signal injection and voltage integration techniques (Combes et al., 2017). With these techniques, a small sinusoidal excitation is applied to the stator windings at low speed or at standstill, and the current response is analysed to extract magnetic and inductive characteristics. Researchers created high-precision models appropriate for controller design and real-time PMSM drive monitoring by combining traditional and advanced testing techniques, especially in electric vehicle applications where stability and efficiency are crucial (Zhu et al., 2021).

However, there is a need for this research because few studies have experimentally validated PMSM parameter identification for small-scale electric drive applications like the Shell Eco Marathon prototype.

Bibliography

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APPENDICES:

Materials and supporting data for the project proposal are included in the following appendices. Additional information is included in each appendix to support the main report

Appendix A: System Overview description

The three-phase inverter that powers the PMSM is powered by a DC power source in the suggested configuration for PMSM parameter identification and control. The motor has an encoder to provide information on the position of the rotor and current and voltage sensors for electrical feedback. A microcontroller that uses the Field-Oriented Control (FOC) algorithm processes these signals. For accurate speed and torque control, the inverter switching is controlled by the controller output. Performance metrics are recorded and outcomes are validated using data acquisition tools.

Appendix B: Bill of materials

Component	Specification	Purpose
PMSM	350W, 48V, 3000rpm	Motor under test
DC Power Supply	0-60V ,10A regulated	Provides power to inverter
Three phase Inverter	IGBT	Controls motor phase via PWM
Microcontroller	Arduino	Implements FOC algorithm
Current Sensor	Hall-effect sensor (20A)	Measures phase current
Voltage Sensors	Divider-type	Measures DC bus voltage
Position sensor	Incremental rotary encoder	Provides rotor position
Oscilloscope	Two-channel digital scope	Records voltage and current waveform
Safety Equipment	Insulated gloves, circuit isolation tools	Ensures lab safety during testing

Appendix C: Key Equations and PMSM Model:

- **Stator Voltage equation:**

$$\boldsymbol{v_d = R_s i_d - \omega_e L_q i_q + d_{\psi_d}/dt}$$

$$v_q = \boldsymbol{R_s i_d + \omega_e L_q i_q + d_{\psi_d}/dt}$$

Where:

v_d, V_q = stator voltages in d–q axes

i_d, i_q = stator currents in d–q axes

R_s = stator resistance

L_d, L_q = d–q axis inductances

$\psi d, \psi q$ = stator flux linkages

ωe = electrical angular speed

Torque Equation:

$$T_e = \frac{3}{2} p[\psi f iq + (Ld - Lq) idiq]$$

Where:

T_e = electromagnetic torque

p = of pole pairs

ψ_f = permanent magnet flux linkage

Appendix D: Ethical and Safety Considerations

All experiments will be carried out in compliance with the laboratory safety procedures established by the School of Engineering at Coventry University. All tests will use the proper personal protective equipment (PPE), such as safety glasses and insulated gloves, because the PMSM test setup involves electrical equipment running at moderate voltage and current levels. Before making any changes to the wiring or configuration, the circuit will be isolated and the power will be disconnected. To reduce the possibility of electric shock or component damage, the setup will only be tested in a laboratory setting under supervision. All measurements and results will be accurately recorded, reported, and used only for academic and research purposes thanks to data collection that complies with ethical standards.